Thakaa at ArA-DF 2026: Fusing Self-Supervised and Signal-Processing Detectors for Arabic Speech Deepfake Detection

Abeer Alhammad **Abdullah Aldahlawi**
Thakaa, Advanced Company for Information Technology
Riyadh, Saudi Arabia
{a.alhammad, aldahlawi}@thakaait.net

Abstract

ArA-DF 2026 is a shared task on binary Arabic speech deepfake detection, with two tracks: dialect generalisation and acoustic robustness. For both we submit rank-normalised score fusions of complementary detectors. Track 1 fuses five components in two stages: three fine-tuned XLS-R-300M networks and two ResNet-18 classifiers over linear-prediction residual and group-delay representations. Track 2 fuses two XLS-R-300M detectors differing in augmentation strategy. We reach 2.8961 EER on Track 1 and 2.7626 on Track 2, against baselines of 14.54 and 27.11. Our analysis on the released test labels shows individual EER to be a poor guide to fusion contribution, which depends on error independence: both fusions improve on their best member by over 40%, and components far weaker in isolation than any we submitted improve them further.

1 Introduction

ArA-DF 2026 (ArA-DF 2026 Organizers, 2026) is a shared task at ArabicNLP 2026 on binary Arabic speech deepfake detection. Track 1, *dialect generalisation*, tests reliability on dialects and speakers absent from training; Track 2, *acoustic robustness*, tests degraded acoustic and channel conditions. Both are ranked by equal error rate (EER).

The task matters because it varies dialect and acoustics independently: in a single pooled test set, failing on an unfamiliar dialect and failing on a degraded recording produce the same symptom. Both problems matter for Arabic: Modern Standard Arabic and the regional dialects differ enough that a detector trained on one variety may not transfer to another, and audio encountered in practice carries compression, background noise and re-recording effects rather than studio conditions.

Our submitted systems reach 2.8961 EER on Track 1 and 2.7626 on Track 2. Our strategy on both tracks was score-level fusion under rank normalisation: we kept components that made *different* mistakes from the system we already had, rather than those with the best individual scores. On Track 1 we pursued that difference architecturally, fusing three fine-tuned transformers (Babu et al., 2022) alongside two classifiers that read processed representations rather than the waveform; on Track 2 we pursued it through training, pairing a component targeted at a measured failure mode with one augmented more generically.

We report three findings. Individual EER was a poor guide to fusion contribution: the fusions improve on their best member by 40% and 44%, and two components at 14.59 and 10.17 EER that we never scored on the evaluation set improve the Track-2 fusion further. Development-set scores did not predict it either: **rev-noise** scored 6.0670 on the held-out development-phase partition, yet removing it from the two-component Track-2 fusion leaves **reverb** alone at 4.9370, a cost of 79%. Finally, the condition we targeted for augmentation proves close to benign on the released labels, at $1.3\times$ the clean error rate, while the worst condition, an elevated noise floor, costs $2.1\times$.

2 Background and Task Setup

2.1 Input and Output

Systems receive single-channel audio utterances and must distinguish bona fide speech from the output of a text-to-speech or voice-conversion system. Submissions are continuous scores, one per utterance, ranked by equal error rate.

2.2 Dataset

The ArA-DF 2026 dataset (ArA-DF 2026 Organizers, 2026) provides 22,500 unique training utterances at 2:1 spoofed to bona fide, balanced across Modern Standard, Gulf and Egyptian; development holds 21,000 at the same class ratio but 71.5% Gulf. The evaluation sets hold 144,210 (Track 1) and

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1507–1511
October 28-29, 2026 ©2026 Association for Computational Linguistics

Model	Tr.	Backbone	Distinguishing feature
base	1	XLS-R	5 augmentation classes
rev-noise	2	XLS-R	9 augmentation classes
reverb	1, 2	XLS-R	room simulation, $p = 0.6$
dial-inv	1	XLS-R	AASIST, $\lambda = 1$
lp-res	1	ResNet-18	LP residual
gd	1	ResNet-18	group delay

Table 1: Component detectors and the track (Tr.) each is used on. The XLS-R models are 1.27 GB; **lp-res** and **gd** are 0.01 GB and use no pretrained model. All XLS-R components carry a gradient-reversed dialect head; **base** and **rev-noise** additionally carry a reversed acoustic-condition head.

127,746 (Track 2); labels were released after the evaluation phase closed and we use them for the post-submission analysis in Section 5. Their class balance is more skewed than the training partition, at roughly 3:1 spoofed. Our combined training index holds 67,500 entries: the 22,500 originals plus two pre-rendered augmented copies of each, carrying an elevated noise floor. Two components extend this with the development partition (Section 4.1).

2.3 Related Work

Detectors built on large self-supervised speech encoders such as XLS-R (Babu et al., 2022) now dominate anti-spoofing, typically by fine-tuning the encoder and reading it with a lightweight graph-attention back-end (Jung et al., 2022). Such systems are commonly fused at the score level, though the components often share that front-end. Phase-based representations predate this convergence and remain complementary to magnitude-spectral features (Sanchez et al., 2015; Yang et al., 2021). Domain-adversarial training (Ganin and Lempitsky, 2015) suppresses nuisance factors in the learned representation, and augmentation with recorded room impulse responses (Ko et al., 2017) addresses conditions absent from training. We fine-tune such an encoder, use both adversarial training and impulse-response augmentation, and add classical representations alongside.

3 System Overview

3.1 Model Architecture

Four of the six components (Table 1) fine-tune XLS-R-300M (Babu et al., 2022): the convolutional front-end is frozen and the upper transformer layers are updated. Three use statistics pooling, concatenating the mean and standard deviation of the frame representations over time; **dial-inv** instead uses an AASIST graph-attention back-end (Jung et al., 2022). Each carries a spoof classifier and a gradient-reversed dialect head (Ganin and Lempitsky, 2015), intended to discourage the encoder from using dialect as a proxy for the spoof decision. **base** and **rev-noise** additionally carry a reversed head over the acoustic condition applied by augmentation — five classes and nine respectively — so that the model cannot infer the spoof label from the type of degradation present. **reverb** and **dial-inv** carry the dialect head only.

Two components use no pretrained model. **lp-res** reads an STFT of the linear-prediction residual, which isolates the excitation once the vocal-tract envelope is removed; **gd** reads a group-delay spectrum, a phase-domain representation that discards magnitude (Sanchez et al., 2015; Yang et al., 2021). Both are ResNet-18 classifiers (He et al., 2016) over four-second windows with log-sum-exp pooling, roughly 1% the size of the transformers.

3.2 Fusion Algorithm

Components are combined at the score level. For component m we replace raw scores by percentile ranks $R_m(i) \in (0, 1]$ over the evaluation set and combine them linearly, $s(i) = \sum_m w_m R_m(i)$ with $\sum_m w_m = 1$. Rank normalisation is what makes the weights meaningful: the components emit scores on incompatible scales, and without it the widest-ranged component dominates regardless of w_m. It is lossless with respect to the metric, since EER depends only on the induced ordering. Track 2 applies this once; Track 1 uses multi-stage fusion (Kurnaz et al., 2025), with **reverb** entering at both stages.

3.3 Design Choices

Score fusion improves on its components only to the extent that they make different errors. Augmentation gives some diversity, since a component weights evidence according to what it was trained to survive, but components sharing an encoder and an input remain correlated. This constraint shaped which components we retained.

Track 1. The five components vary along three dimensions — objective, training regime and input representation — with **base** as the reference. **dial-inv** varies the objective, penalising dialect recoverability three times more strongly and replacing statistics pooling with attention over frames.

reverb varies the training regime, updating six transformer layers rather than twelve under room-simulation augmentation. **lp-res** and **gd** vary the input itself, discarding the spectral envelope and the magnitude spectrum respectively, and so read the signal on a different axis from the transformers.

Track 2. Both components read the waveform through XLS-R, so we varied augmentation and fine-tuning depth: **rev-noise** was augmented against the degradation combination we had measured as most costly and updates twelve transformer layers; **reverb**, the same component used in Track 1, applies room simulation and updates six transformer layers.

3.4 External Resources

We use the pretrained XLS-R-300M encoder (`facebook/wav2vec2-xls-r-300m`) (Babu et al., 2022), the AASIST back-end architecture (Jung et al., 2022) for **dial-inv**, and the OpenSLR-28 room impulse response and noise corpus (Ko et al., 2017) for augmentation. No external training audio or additional pretrained models were used.

3.5 Submitted Systems

Track 1. A multi-stage fusion. The inner stage combines all five components at weights 0.3645, 0.1869, 0.1589, 0.1495 and 0.1402 for **base**, **reverb**, **dial-inv**, **lp-res** and **gd** respectively, giving an intermediate score p. The outer stage re-ranks p and fuses it with **reverb**, which therefore enters twice:

$$s = 0.60\,R(p) + 0.40\,R_{\text{rev}}. \tag{1}$$

The inner weights were set by a weight search; the outer weighting was selected by comparing leaderboard scores across submissions.

Track 2. A single-stage fusion of **rev-noise** at 0.35 and **reverb** at 0.65, the weighting again selected by comparing leaderboard scores.

4 Experimental Setup

4.1 Train, Development and Test Usage

All components train on the 67,500-entry index of Section 2.2. **base** and **rev-noise** additionally concatenate the development partition into training and select checkpoints on that same data, so their training-time development EER is not held out; the remaining four train on the training partition only, which we verified excludes the development partition. Each track has its own unlabelled development-phase partition — 16,023 utterances on Track 1 and 14,193 on Track 2, both disjoint from the 21,000 labelled split — so scores on them remain held out for all components. Every development-phase figure we report was scored on the Track 2 partition. No component trains on the evaluation partition.

4.2 Preprocessing

Audio is resampled to 16 kHz and cropped to four seconds. Eight- and twelve-second windows scored 3.9598 and 3.6204 on the development phase, against 3.3894 at four seconds; neither is submitted. **rev-noise**, **base** and **dial-inv** take a random crop offset during training and a centre crop at evaluation; **reverb** takes the leading four seconds throughout. The two signal-processing components analyse the full utterance. All peak-normalise. At inference each utterance is scored in a single forward pass; the transformers use no multi-crop averaging.

4.3 Augmentation

Augmentation is applied on the fly, in addition to the pre-rendered copies in the training index. **base**, **rev-noise** and **dial-inv** share a shortcut-breaking step, which for **dial-inv** is the whole of its augmentation: the spoofed class carried more digital silence and a quieter noise floor than bona fide speech, so digital zeros are filled, DC offset and floor randomised identically for both classes, and rooms applied at probability 0.30. **rev-noise** then draws from nine acoustic classes: a clean pass, five single degradations — reverberation, events, sustained noise, channel simulation and band-limiting — and three combinations, with 34% of draws on reverberation-plus-noise; its noise is sampled from OpenSLR-28 (Ko et al., 2017). **base** uses a five-class version of the same scheme. **reverb** applies room simulation at probability 0.6 with procedurally synthesised background and no shortcut-breaking step. Impulse responses come from the same corpus. **lp-res** and **gd** use the pre-rendered copies only. We verified the acoustic modules by instrumenting the samplers and counting draws, confirming the configured priors.

4.4 Hyperparameters

All transformer components use AdamW at learning rate 10^{-5} with $\lambda = 0.3$, except **dial-inv** at $\lambda = 1$ and 10^{-4}. **base** and **rev-noise** train for 14 epochs at batch size 8; **reverb** for 12 epochs at

batch size 16; **dial-inv** for 10 epochs at batch size 4. **lp-res** and **gd** train for 7 and 8 epochs at batch sizes 24 and 32, learning rate 3×10^{-4}. Checkpoints are selected by lowest development EER, which for **dial-inv**, **lp-res** and **gd** was the final epoch trained.

4.5 Software and Evaluation

PyTorch 2.1.0 (CUDA 12.1), `transformers` 4.35.2, `torchaudio` 2.1.0 and `librosa` 0.11.0, on NVIDIA RTX A6000 GPUs. Systems are ranked by equal error rate. All reported figures come from single training runs rather than seed averages.

5 Results and Analysis

5.1 Official Results and Ranking

Our submitted fusions reach 2.8961 EER on Track 1 and 2.7626 on Track 2, against organiser baselines of 14.54 and 27.11, placing ninth and sixth respectively.

5.2 Ablations

Single-component submissions (official). Submitting **reverb** alone, without the other components, scores 4.8423 on Track 1 and 4.9370 on Track 2 — costs of 67% and 79%: it is valuable in combination but weak in isolation.

Fusion against its components (post-submission). The evaluation labels are used below for measurement only; we reproduced both submitted systems exactly from their component scores. Individually the Track-1 components span 4.8423 to 10.4923 against 2.8961 fused; on Track 2, 4.9370 (**reverb**) and 5.8047 (**rev-noise**) against 2.7626. **gd** and **base**, both Track-1 components, were also scored on the Track-2 development-phase partition, the only set held out from every component, and ranked differently there: **gd** scored best, at 3.3894, yet is second-worst on the Track-1 evaluation set and carries the smallest inner weight, while **base**, worst at 6.5290, carries the largest; **rev-noise** scored 6.0670 there. Correlations point the same way: independence, not EER, is what earns weight. On Track 2 the pair correlates at 0.554, with only 27% error overlap. On Track 1 correlations with the inner fusion span 0.847 to 0.933, with **reverb** least correlated, and 34–49% of each component's errors are unique.

Unsubmitted components (post-submission). Of several components never scored on the evaluation set, two use frozen HuBERT and Whisper encoders with light classifiers. Both are far weaker in isolation than any submitted component, at 14.59 and 10.17 EER, yet overlap the Track-2 submitted pair's errors at only 9.0% and 5.3%, where the submitted components themselves overlap it at 31.6% and 44.5%. Adding either at weights between 0.05 and 0.15 improves the fusion to 2.6509 and 2.6604 on the released labels; adding both reaches 2.6221.

Condition	% of set	EER	vs. clean
silence-heavy	4.9%	1.46	0.6×
event	6.6%	2.00	0.8×
clean	57.6%	2.47	—
reverberant	21.9%	3.09	1.3×
low SNR	9.0%	5.15	2.1×

Table 2: Track-2 EER (%) by dominant acoustic condition, classified by our own signal-based analysis over all 127,746 evaluation utterances. Post-submission.

5.3 Error Analysis

By acoustic condition. Classifying the Track-2 evaluation audio by dominant degradation (Table 2) shows error is not concentrated where we had aimed. An elevated noise floor is the worst condition at 2.1× the clean rate despite covering only 9.0%, while reverberation costs 1.3×. Silence-heavy audio scores best of any condition and is 89% spoofed, matching the training-set association that the shortcut-breaking augmentation targets.

Confusion pattern. At a threshold matched to the evaluation prior, the systems reach 98.2% accuracy on spoofed speech and 94.8% on bona fide: the dominant failure is a false accusation rather than a missed spoof.

Scope. Our error analysis is necessarily confined to acoustics, since dialect labels are not part of the released evaluation metadata.

6 Conclusion

We described rank-normalised score fusions for both ArA-DF 2026 tracks, reaching 2.8961 and 2.7626 EER — more than 40% better than the best single component. Offline measurement guided us poorly: development ranking inverted against fusion weight, and the combination we targeted proved benign while noise alone, which we had treated only in combination, proved worst. Future work will retarget augmentation at the noise floor alone.

Limitations

All figures come from single training runs rather than seed averages, so small differences between components may not be stable. Our fusion weights were selected from development-leaderboard feedback, the only signal available before the evaluation labels were released; refitting on those labels improves Track 1 by 0.13 EER and Track 2 not at all. **base** and **rev-noise** select checkpoints on data they also train on, so their development EER is not held out. Every analysis using the released evaluation labels is exploratory: the unsubmitted-component weights in Section 5 were chosen with those labels visible, so 2.6221 is an upper bound rather than a blind score. Finally, the released metadata carries no dialect field, so no dialect-level error analysis was possible on Track 1.

Ethical Considerations

Deepfake detection is dual-use: the error analysis in Section 5 identifies which acoustic conditions defeat our system, and could guide an attacker toward them.

Our two error types carry different consequences. A missed spoof leaves a synthetic recording unchallenged; a false accusation attributes a fabrication to a real speaker. Our systems fail predominantly in the second direction, misclassifying 5.2% of bona fide utterances against 1.8% of spoofed ones at the prior-matched threshold used for analysis. Any deployment should set the operating point according to which error is more costly in that setting.

The dataset covers Egyptian, Gulf and Modern Standard Arabic; performance on other varieties is untested, and the released evaluation metadata carries no dialect field, so per-dialect error rates cannot be measured even on the varieties present. We used only the audio released by the organisers.

Acknowledgments

We thank the ArA-DF 2026 organisers for the shared task, and Thakaa for supporting this work.

References

ArA-DF 2026 Organizers. 2026. ArA-DF 2026: Arabic speech deepfake detection shared task. https://huggingface.co/datasets/ArabicSpeech/ArA-DF-2026. ArabicNLP 2026 Shared Task 14.

Arun Babu, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, Alexei Baevski, Alexis Conneau, and Michael Auli. 2022. XLS-R: Self-supervised cross-lingual speech representation learning at scale. In *Proc. Interspeech*, pages 2278–2282.

Yaroslav Ganin and Victor Lempitsky. 2015. Unsupervised domain adaptation by backpropagation. In *Proc. ICML*, pages 1180–1189.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proc. CVPR*, pages 770–778.

Jee-weon Jung, Hee-Soo Heo, Hemlata Tak, Hye-jin Shim, Joon Son Chung, Bong-Jin Lee, Ha-Jin Yu, and Nicholas Evans. 2022. AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks. In *Proc. ICASSP*, pages 6367–6371.

Tom Ko, Vijayaditya Peddinti, Daniel Povey, Michael L. Seltzer, and Sanjeev Khudanpur. 2017. A study on data augmentation of reverberant speech for robust speech recognition. In *Proc. ICASSP*, pages 5220–5224.

Oğuzhan Kurnaz, Tomi H. Kinnunen, and Cemal Hanilçi. 2025. Investigating the potential of multi-stage score fusion in spoofing-aware speaker verification. In *Proc. SIU*, pages 1–4.

Jon Sanchez, Ibon Saratxaga, Inma Hernáez, Eva Navas, Daniel Erro, and Tuomo Raitio. 2015. Toward a universal synthetic speech spoofing detection using phase information. *IEEE Transactions on Information Forensics and Security*, 10(4):810–820.

Jichen Yang, Hongji Wang, Rohan Kumar Das, and Yanmin Qian. 2021. Modified magnitude-phase spectrum information for spoofing detection. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:1065–1078.